

Race/Ethnicity, Sex, and Socioeconomic Status as Predictors of Outcome After Surgery for Temporal Lobe Epilepsy

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Background: Several risk factors have been attributed to seizure recurrence after surgery. It is unknown whether race/ethnicity plays a role in outcome.

Objective: To evaluate whether race/ethnicity plays a role in seizure recurrence after surgery.

Design: Cohort study.

Setting: We evaluated data obtained from the epilepsy centers at the University of Alabama at Birmingham and New York University, New York, NY.

Patients: All patients included had a diagnosis of mesial temporal sclerosis and underwent temporal lobectomy.

Main Outcome Measures: Occurrence of seizure after surgery was registered 1 year after surgery. We used multiple logistic regression analysis to model the presence of seizure recurrence after surgery and generated odds ratios (ORs) for seizure recurrence after surgery for African American and Hispanic patients relative to white patients. An unadjusted model incorporated only race/

ethnicity as the independent variable, and an adjusted model included socioeconomic status, age, duration of epilepsy, education, history of febrile seizures, sex, handedness, lateralization of epileptogenic focus, and number of antiepileptics as the independent variables.

Results: Two hundred fifty-two patients underwent surgical treatment with pathological confirmation of mesial temporal sclerosis. No differences were found between racial/ethnic groups in terms of seizure recurrence in any models. For African American patients, the ORs were 0.9 (95% confidence interval [CI], 0.4-2.1) for the unadjusted model and 0.8 (95% CI, 0.3-2.0) for the adjusted model; for Hispanic patients, the ORs were 1.6 (95% CI, 0.8-3.2) for the unadjusted model and 1.1 (95% CI, 0.5-2.6) for the adjusted model, relative to white patients.

Conclusion: Our data suggest that although sex appears to play a role in the outcomes of surgery for temporal lobe epilepsy, race and socioeconomic status do not.

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THE SUCCESS OF SURGERY IN temporal lobe epilepsy (TLE) associated with mesial temporal sclerosis (MTS) is higher than that for other types of epilepsy, reaching 58% for patients who were free of seizures impairing awareness, when compared with medical treatment, based on a randomized controlled trial.¹ There are, however, marked racial/ethnic disparities in the use of this effective treatment. African American patients have lower rates of surgical intervention for treatment of MTS compared with white patients,² whereas Hispanic patients in some areas of the United States do not have any access.² The lack of access to surgical treatment in developing countries is an even greater challenge.³

The reasons for the disparities remain unclear. African American patients may experience a lower psychosocial impact of epilepsy^{4,5} and may have a preference for non-surgical management of chronic problems. However, physicians' perceptions and expectations of patients can vary according to factors such as race, which could influence referrals for TLE surgery.⁶ Furthermore, the lack of patient education that results from these erroneous perceptions is a serious problem. It is critical that we have information regarding treatment outcomes in different racial/ethnic groups so that we can better inform our patients with evidence-based data.

An initial look into this problem in a small group of patients revealed that race had an effect on seizure outcome after temporal lobectomy, with African American

patients having a significantly higher recurrence rate than white patients.⁷ However, in that study, the sample size was small, and other minorities such as Hispanic or Latino patients were not included, although they constitute the largest minority population in the United States.^{7,8}

In the present study, we examined racial/ethnic differences in outcome among a large cohort of patients with medically intractable mesial TLE who underwent temporal lobectomy. We also examined potential modifying variables among racial groups.

METHODS

Patient-specific discharge and surgical data were obtained from the videoelectroencephalographic units at the University of Alabama at Birmingham and New York University, New York, NY, from patients released with the primary diagnosis of TLE from July 1, 1994, through January 31, 2003. The data set includes information on each patient's age, ZIP code of residence, date of birth, sex, race/ethnicity, date of surgery, degree of education, duration of epilepsy (in years), handedness, lateralization of temporal epileptogenic focus, number of antiepileptics used before surgery, and history of febrile seizures.

SAMPLE

All patients discharged with a primary diagnosis of medically intractable TLE were identified. The diagnosis was based on the confirmation of an ictal onset in either temporal region by means of videoelectroencephalographic evaluation, temporal lobe interictal epileptiform abnormalities, clinical semiology consistent with TLE, and intracranial electroencephalographic confirmation of mesial temporal onset in a subset of patients.

All patients without a magnetic resonance imaging (MRI) diagnosis of hippocampal sclerosis or with another pathology associated with hippocampal sclerosis (in addition to MTS) were excluded. The final group consisted of patients with MTS who received a temporal lobe resection.

MAGNETIC RESONANCE IMAGING

All patients underwent imaging on a 1.5-T MRI system using standardized protocols. We acquired sagittal and coronal T1-weighted, coronal T2-weighted spin echo, fluid-attenuated inversion recovery, and inversion recovery sequences. We diagnosed MTS on the basis of evidence of hippocampal atrophy and abnormal signal alterations according to validated diagnostic criteria.^{9,10}

SURGERY AND OUTCOME

All patients who decided to undergo surgery for TLE due to hippocampal sclerosis received a resection of a portion of neocortex and a targeted resection of the amygdala and hippocampus (including the anterior two thirds or the entire hippocampus). Patients underwent regular assessment for postoperative seizure control by the treating neurologist in periodic follow-up visits in their respective epilepsy clinic. All patients had at least 1 year of follow-up. Outcome at the most recent follow-up was classified as seizure free or not seizure free. In the seizure-free group, we also included patients who still had auras (Engel I), including in the immediate postsurgical period.

PATHOLOGICAL ANALYSIS

Representative tissue samples of the hippocampus were routinely obtained for analysis. Specimens were fixed in formalin

and embedded in paraffin. Histologic sections of the hippocampus and temporal lobe were stained with hematoxylin-eosin and glial fibrillary acidic protein. Adequate hippocampal tissue was available for the diagnosis of hippocampal sclerosis in most subjects. We diagnosed MTS according to the presence of neuronal loss in the cornu ammonis (CA) 1 and CA3-4 regions of the hippocampus by qualitative criteria. If the pathological findings were not diagnostic for MTS (inadequate sample), the patient was included on the basis of MRI identification of MTS.

STUDY VARIABLES

Independent variables consisted of race/ethnicity, age, sex, income, duration of epilepsy, education, history of febrile seizures, handedness, lateralization of temporal epileptogenic focus, and number of antiepileptics used. Median household annual income by residence ZIP codes obtained from the 2000 US census was used as a proxy for patient income.¹¹⁻¹⁵ The following 3 specific ethnic groups constituted the race/ethnicity variable: African American, Hispanic or Latino, and white. Ethnicity was registered by the treating physician, always corroborating with the patient's last name. The following 4 specific age categories were used: younger than 25, 25 to 39, 40 to 49, and 50 years or older. Income was grouped into 4 categories, based on average annual income in US dollars: less than \$35 000, \$35 000 to \$54 999, \$55 000 to \$74 999, and more than \$74 999. Education was grouped as primary and secondary school, college, graduate and postgraduate training, and unknown. Finally, antiepileptics were grouped as 1, 2, and more than 2 antiepileptics.

ANALYSIS METHODS

Multivariate logistic regression was used to investigate the relationship between seizure recurrence and race/ethnicity, with potential confounders controlled. Two sets of logistic regression models were estimated to generate odds ratios (ORs) for seizure recurrence after surgery for African American and Hispanic patients relative to white patients. The first model incorporated only ethnicity as the independent variable and generated unadjusted ORs for having seizure recurrence. The second set included all independent variables described in the preceding subsection. Statistics were computed with SAS version 9.0 software (SAS Institute Inc, Cary, NC) for Windows (Microsoft Corp, Redmond, Wash).

RESULTS

Two hundred fifty-two patients underwent surgical treatment with pathological or MRI confirmation of MTS. Their mean (SD) age was 35.5 (11.9) years. Twenty-six patients were African American, 40 were Hispanic, and 186 were white. Men represented 49.6% of the sample; 86.5% were right-handed; 69.8% had febrile seizures; and 44.0% had left-sided TLE. Seizure recurrence was seen in 53.2% (**Table 1**).

All independent variables potentially associated with seizure recurrence among racial/ethnic groups were examined (Table 1). There were no differences between the groups with respect to all the variables, with the exception of sex, history of febrile seizures, and socioeconomic status.

Logistic regression results are shown in Table 2. Model 1 includes unadjusted ORs for African American and His-

Table 1. Distribution of Independent Variables Assessed in Patients Who Underwent Temporal Lobectomy for Mesial Temporal Sclerosis*

Variables	Non-Hispanic White	African American	Hispanic or Latino
Seizure recurrence	96	13	25
Socioeconomic status level†			
1	28	8	21
2	81	15	14
3	58	2	4
4	22	1	1
Age at surgery, y			
<25	38	1	8
25-39	74	16	22
40-49	51	6	7
≥50	23	3	3
Duration of epilepsy, y			
<10	28	7	6
10-30	93	14	25
>30	65	5	9
Education			
Primary or secondary school	88	12	21
College	38	6	9
Graduate and postgraduate training	44	4	5
Unknown	16	4	5
Febrile seizures	122	23	31
Sex			
Male	82	17	26
Female	104	9	14
Handedness			
Right	162	23	33
Left	21	3	5
Ambidextrous	3	0	2
Right temporal lobectomy	106	17	18
No. of antiepileptics			
1	23	2	10
2	84	8	15
>2	79	16	15

*Data are expressed as number of patients.

†Level 1 indicates an annual income (in US dollars) of less than \$35 000; 2, \$35 000 to \$54 999; 3, \$55 000 to \$74 999; and 4, more than \$74 999.

panic patients, with white patients as the reference group. Hispanic patients were more likely to have seizure recurrence before adjustments were made (OR, 1.6), although the 95% confidence interval included the null value.

In the multivariate logistic regression analysis that controlled for all independent variables (model 2), ORs for African American and Hispanic patients indicated no significant difference when compared with white patients because the 95% confidence interval included the null value.

Finally, the analysis demonstrated that male sex was associated with a reduced risk of seizure recurrence (OR, 0.5; 95% confidence interval, 0.3-0.8) (**Table 2**).

COMMENT

Using prospectively collected data, we found that seizure outcomes at least 1 year after temporal lobectomy

Table 2. Summary of Logistic Regression Analyses*

Variables	OR (95% CI)	
	Unadjusted†	Adjusted‡
Race/ethnicity		
Non-Hispanic white	1.0	
African American	0.9 (0.4-2.1)	0.8 (0.3-2.0)
Hispanic or Latino	1.6 (0.8-3.2)	1.1 (0.5-2.6)
Duration of epilepsy, y		
<10	1.0	
10-30	1.6 (0.8-3.2)	1.4 (0.6-3.2)
>30	1.5 (0.7-3.3)	1.7 (0.7-4.5)
Education		
Primary or secondary school	1.0	
College	1.5 (0.8-2.8)	1.6 (0.9-3.2)
Graduate or postgraduate training	0.4 (0.1-2.4)	0.5 (0.1-2.9)
Unknown	0.4 (0.1-1.0)	0.3 (0.1-0.9)
Febrile seizures	1.0 (0.6-1.8)	0.8 (0.5-1.6)
Male sex	0.6 (0.4-1.0)	0.5 (0.3-0.8)
Side of surgery		
Right	1.0	
Left	0.9 (0.6-1.5)	0.8 (0.5-1.5)
Handedness		
Right	1.0	
Left	0.9 (0.4-2.0)	0.9 (0.4-2.2)
No. of antiepileptics		
1	1.0	
2	1.0 (0.5-2.2)	1.0 (0.4-2.2)
>2	0.9 (0.4-1.9)	0.6 (0.3-1.5)
Age at surgery, y		
<25	1.0	
25-39	1.5 (0.7-2.9)	1.2 (0.6-2.8)
40-49	0.7 (0.4-1.6)	0.6 (0.2-1.4)
≥50	0.9 (0.4-2.3)	0.8 (0.2-2.4)
Socioeconomic status level§		
1	1.0	
2	0.9 (0.4-1.7)	0.9 (0.4-1.8)
3	0.8 (0.4-1.6)	0.8 (0.3-1.9)
4	0.6 (0.2-1.7)	0.7 (0.2-2.2)

Abbreviations: CI, confidence interval; OR, odds ratio.

*Odds of seizure recurrence are presented relative to white patients.

†No additional independent variables are used.

‡Adjusted for race, age, duration of epilepsy, education, history of febrile seizures, sex, side of surgery, handedness, number of antiepileptics, and socioeconomic status.

§Level 1 indicates an annual income (in US dollars) of less than \$35 000; 2, \$35 000 to \$54 999; 3, \$55 000 to \$74 999; and 4, more than \$74 999.

for mesial TLE were similar among white, African American, and Hispanic patients. We previously found lower rates of epilepsy surgery in African American patients compared with white patients, and although a clear explanation was not found, this was not related to access to health care in the African American group.²

Subsequently, in another publication,⁷ we found a trend for higher rates of seizure recurrence in African American compared with white patients. However, that study was limited by the small sample size and the retrospective design. In the present study, we did not include access to medical insurance as an independent variable in the analysis because it is more probably related to the decision to undergo surgical treatment for intractable TLE.

Furthermore, because the number of African American patients was very small, we could not rule out the

possibility of ascertainment bias for those undergoing surgery. The present study has a larger sample with patients from 2 different medical centers and different geographic locations. Also, it includes a third group, Hispanic or Latino patients, the size of which has tripled in certain states such as Alabama, and which has a large representation in the New York area and has become the largest minority population in the United States.¹² To our knowledge, most studies examining racial differences in neurological diseases have focused on the African American population, and ours is the first large series (to our knowledge) to examine the outcome of epilepsy surgery in Hispanic patients.

Potential limitations of this study include differences in the patient groups, limited geographic sampling, a small sample size for 1 ethnic group, and the methodology for ethnic classification. First, there were some significant differences in the baseline characteristics of the patients, such as socioeconomic status, but this may be an effect of the sample size. Second, although the patient population consisted of white, African American, and Hispanic patients and the information came from 2 geographically different epilepsy centers, it is not representative of the general population because other racial/ethnic groups such as Asians were not included. Third, the relatively small number of African American patients limits the power to detect differences involving subgroups within that group. Fourth, the potential misclassification of race/ethnicity remains an issue, particularly with Hispanic patients; although the name origin can help with this, reports of race/ethnicity can be subjective and are infrequently validated.¹⁶ This important consideration is not unique to our analysis.¹⁷ Nonetheless, it is unlikely that systematic underreporting or overreporting of a particular group is correlated with the propensity to perform invasive treatments like surgery. Fifth, the study was restricted to patients with MTS and intractable TLE. However, although racial differences for other types of epilepsy may not be inferred from this study, the homogeneous population (MTS only) is a major strength. Finally, detailed data on neurosurgeon characteristics (eg, experience with the procedure, individual surgeon volume, and technique) may have influenced the rate found here in a particular direction. However, any effect would be minimal unless large intraindividual racial differences were present among the surgeons.

As with other medical conditions, patients of racial/ethnic minorities have concerns about surgical outcomes and adverse events after surgical procedures, and these concerns can influence patients' willingness to consider the treatment.^{18,19} This may partly account for the disparities observed in epilepsy.

We need to examine race/ethnicity subgroup data when possible (especially when previous work suggests significant differences by race or ethnicity), as recommended by the National Institutes of Health.²⁰ Current evidence-based knowledge does not allow us to know the potential relationship between race/ethnicity and trial outcomes, and this lack of understanding constitutes a barrier to using results of clinical research to address racial/ethnic disparities in health.²¹

Our findings of a sex difference in outcome (worse outcome in women compared with men with TLE) do not have a clear explanation. Studies found that male animals were more prone to seizure activity than were female animals.^{22,23} In humans, men may be more susceptible to temporal lobe–like seizures because of high levels of testosterone²⁴ and more vulnerable to seizure-associated brain damage²⁵ and may have a slightly higher overall incidence of epilepsy when compared with women,^{26,27} although this may not be the case in the localization-related epilepsies.²⁸ On the other hand, greater plasticity has been found in women when compared with men, in terms of memory, in response to the onset of epilepsy.²⁹ Sex differences in seizure recurrence after surgical treatment of medically refractory epilepsy have not been shown to be a predictive factor, but most of the previous analyses^{7,30-32} have included smaller samples than our study did, or simply sex was not considered as a variable under study.

In conclusion, race/ethnicity and socioeconomic status do not appear to play an important role in TLE surgery outcome, and being male appears to have a protective effect in terms of seizure recurrence.

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REFERENCES

- Wiebe S, Blume WT, Girvin JP, Eliasziw M. A randomized, controlled trial of surgery for temporal-lobe epilepsy. *N Engl J Med*. 2001;345:311-318.
- Burneo JG, Black L, Knowlton RC, Faught E, Morawetz R, Kuzniecky RI. Racial disparities in the use of surgical treatment for intractable temporal lobe epilepsy. *Neurology*. 2005;64:50-54.
- Burneo JG, Tellez-Zenteno J, Wiebe S. Understanding the burden of epilepsy in Latin America: a systematic review of its prevalence and incidence. *Epilepsy Res*. 2005;66:63-74.
- Griggs JJ, Engel J Jr. Epilepsy surgery and the racial divide. *Neurology*. 2005;64:8-9.
- Gehlert S, DiFrancesco A, Chang CH. Black-white differences in the psychosocial outcomes of epilepsy. *Epilepsy Res*. 2000;42:63-73.
- van Ryn M, Burke J. The effect of patient race and socio-economic status on physicians' perceptions of patients. *Soc Sci Med*. 2000;50:813-828.
- Burneo JG, Knowlton RC, Martin R, Faught RE, Kuzniecky RI. Race/ethnicity: a predictor of temporal lobe epilepsy surgery outcome? *Epilepsy Behav*. 2005;7:486-490.
- Lieu TA, Newacheck PW, McManus MA. Race, ethnicity, and access to ambulatory care among US adolescents. *Am J Public Health*. 1993;83:960-965.

9. Berkovic SF, Andermann F, Olivier A, et al. Hippocampal sclerosis in temporal lobe epilepsy demonstrated by magnetic resonance imaging. *Ann Neurol*. 1991; 29:175-182.
10. Jackson GD. The diagnosis of hippocampal sclerosis: other techniques. *Magn Reson Imaging*. 1995;13:1081-1093.
11. Soobader M, LeClere FB, Hadden W, Maury B. Using aggregate geographic data to proxy individual socioeconomic status: does size matter? *Am J Public Health*. 2001;91:632-636.
12. 2000 US census data. US Census Bureau Web site. <http://factfinder.census.gov>. Accessed December 20, 2005.
13. Danesh J, Gault S, Semmence J, Appleby P, Peto R. Postcodes as useful markers of social class: population based study in 26 000 British households. *BMJ*. 1999;318:843-844.
14. Sin DD, Svenson LW, Man SF. Do area-based markers of poverty accurately measure personal poverty? *Can J Public Health*. 2001;92:184-187.
15. Krieger N, Waterman P, Chen JT, Soobader MJ, Subramanian SV, Carson R. ZIP code caveat: bias due to spatiotemporal mismatches between ZIP codes and US census-defined geographic areas: the Public Health Disparities Geocoding Project. *Am J Public Health*. 2002;92:1100-1102.
16. Fustinoni O, Biller J. Ethnicity and stroke: beware of the fallacies. *Stroke*. 2000; 31:1013-1015.
17. LaVeist TA. Beyond dummy variables and sample selection: what health services researchers ought to know about race as a variable. *Health Serv Res*. 1994; 29:1-16.
18. Ibrahim SA, Siminoff LA, Burant CJ, Kwok CK. Differences in expectations of outcome mediate African American/white patient differences in "willingness" to consider joint replacement. *Arthritis Rheum*. 2002;46:2429-2435.
19. Ibrahim SA, Stone RA, Han X, et al. Racial/ethnic differences in surgical outcomes in veterans following knee or hip arthroplasty. *Arthritis Rheum*. 2005; 52:3143-3151.
20. National Institutes of Health. NIH policy and guidelines on the inclusion of women and minorities as subjects in clinical research. NIH Office of Extramural Research Web site. http://grants2.nih.gov/grants/funding/women_min/guidelines_amended_10_2001.htm. Accessed October 12, 2005.
21. Burneo JG, Martin R. Reporting race/ethnicity in epilepsy clinical trials. *Epilepsy Behav*. 2004;5:743-745.
22. Standley CA, Mason BA, Cotton DB. Differential regulation of seizure activity in the hippocampus of male and female rats. *Am J Obstet Gynecol*. 1995;173: 1160-1165.
23. Lemmens EM, Lubbers T, Schijns OE, Beuls EA, Hoogland G. Gender differences in febrile seizure-induced proliferation and survival in the rat dentate gyrus. *Epilepsia*. 2005;46:1603-1612.
24. Mejias-Aponte CA, Jimenez-Rivera CA, Segarra AC. Sex differences in models of temporal lobe epilepsy: role of testosterone. *Brain Res*. 2002;944:210-218.
25. Briellmann RS, Berkovic SF, Jackson GD. Men may be more vulnerable to seizure-associated brain damage. *Neurology*. 2000;55:1479-1485.
26. Hauser WA, Annegers JF, Kurland LT. Incidence of epilepsy and unprovoked seizures in Rochester, Minnesota: 1935-1984. *Epilepsia*. 1993;34:453-468.
27. Kotsopoulos IA, van Merode T, Kessels FG, de Krom MC, Knottnerus JA. Systematic review and meta-analysis of incidence studies of epilepsy and unprovoked seizures. *Epilepsia*. 2002;43:1402-1409.
28. Christensen J, Kjeldsen MJ, Andersen H, Friis ML, Sidenius P. Gender differences in epilepsy. *Epilepsia*. 2005;46:956-960.
29. Helmstaedt C, Brosch T, Kurthen M, Elger CE. The impact of sex and language dominance on material-specific memory before and after left temporal lobe surgery. *Brain*. 2004;127:1518-1525.
30. Spencer SS, Berg AT, Vickrey BG, et al. Predicting long-term seizure outcome after resective epilepsy surgery: the multicenter study. *Neurology*. 2005;65: 912-918.
31. Jutila L, Immonen A, Mervaala E, et al. Long term outcome of temporal lobe epilepsy surgery: analyses of 140 consecutive patients. *J Neurol Neurosurg Psychiatry*. 2002;73:486-494.
32. Janszky J, Janszky I, Schulz R, et al. Temporal lobe epilepsy with hippocampal sclerosis: predictors for long-term surgical outcome. *Brain*. 2005;128:395-404.

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